



BATKA: batch authentication and tree-based key agreement for secure and efficient grouping of vehicles in 5G-VANET

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Received: 1 October 2024 / Accepted: 28 April 2025 / Published online: 11 December 2025
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Abstract

With the advancement and popularization of the Internet of Vehicles technology, we can add many functions to the vehicle, which promotes the transformation of the vehicle into an intelligent vehicle. Intelligent vehicles are a part of the Internet of Everything (IoE). However, the IoE situation results in a massive surge in the number of connected devices, making it possible for hundreds of pieces of information to be sent simultaneously to the roadside unit (RSU) or gNB from devices related to wireless communication. The traditional single-message authentication process incurs substantial computational overhead, making it less efficient for large-scale deployments. We propose a group scheme based on elliptic curve cryptography (ECC) batch authentication combined with a tree-based key structure to satisfy these requirements. The scheme consists of two parts. The first part abandons the problem of traditional vehicular ad-hoc network's certificate center management. It allows batch authentication, thus reducing overhead and improving efficiency. The second part introduces the tree-based key agreement structure. It enables multiple vehicles to form a group through the group leader. Group members can not only authenticate each other as members of the same group but also have a key corresponding to the encryption key, which can be used to transmit information within the group. Each time a new member joins the group, it must be authenticated. Finally, we show that our proposed scheme is safe and efficient through security analysis and simulation results.

Keywords Batch Authentication · Group · Vehicular ad hoc networks · Privacy · Tree-based Key agreement

1 Introduction

5G mobile communication network plays an essential role in wireless communication transmission of the Internet of Vehicles (IoV). With the advancement of this technology, the Vehicular ad-hoc network (VANET) has also risen rapidly. Many papers related to and combined with these two technologies have been published [1–3]. In this chapter, we

present the relevant background knowledge, followed by the motivation and the contribution of our research.

1.1 Background

In recent years, many devices have been moving toward unmanned automation to make people's lives more convenient and prevent them from being affected by unexpected events. Furthermore, with the advancement of battery storage technology and the rise of environmental awareness, the new generation of vehicles represents the return of electric vehicles to the cutting-edge trend of the times from the idea that was eliminated in the original history. The electric vehicle crystallizes many modern technologies. The interconnection between smart vehicles enables the exchange of information, such as forecasting severe weather conditions or traffic conditions, allowing drivers and passengers a safer, more comfortable, and more convenient ride experience, and promoting the rapid development of in-vehicle communication technology [4–7].

As the predecessor of IoV, VANET can effectively combine information about drivers, vehicle conditions, and road

This article is part of the Topical Collection: 4 - *Track on IoT*
Guest Editor: Peter Langendoerfer

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conditions [8] and provide drivers with other conditions outside the visual range, helping to reduce blind spots and improve traffic efficiency [9]. At the same time, VANET is based on the principle of MANET and applied to the vehicle field technology. The concept is that the vehicle is a node in the network. The devices that can communicate with the vehicle are also regarded as nodes. The nodes transmit information and communicate to form a communication network. Moreover, people use this association to think about combining the existing 5G mobile network with the IoV.

With the quick development of mobile networks, the fifth generation of mobile networks (5G) has become popular. Because of the outstanding technical characteristics of the 5G network, such as lower delay, high transmission rate, and high reliability, etc., and the emerging communication technologies such as millimeter wave communication, MIMO, D2D, and other technologies, through the combination of 5G network technology and VANET [10], can overcome the limitations of the current IoV, bring a historic breakthrough to the IoV [11, 12], and also make the integration of 5G technology and IoV attract more attention [13].

To achieve safety applications that can decrease personal injuries arising from traffic accidents or avoid vehicle collisions effectively [14], the current research on the IoV mainly concentrates on the safety of driving, vehicle traffic efficiency improvement, and both safety and efficiency in communication between vehicles and between vehicles and infrastructure. The existing communication of the in-vehicle network mostly counts on Cellular-based Vehicle-to-Everything (C-V2X). Safety during automatic driving can be ensured via communication between vehicles and others (such as people, infrastructure, and networks) [15]. C-V2X includes two standards: LTE-V2X and 5G-V2X. Comparing the two standards, because of the characteristics of 5G [16], the performance of 5G-V2X is more outstanding than LTE-V2X. The advantage of 5G-V2X is low latency, extended coverage time, and high reliability.

5G-V2X can receive diverse real-time road status information accurately and interact with others in time. It can also achieve the task of driving better. To implement V2X communication, multiple protocols are adopted in IoV, including cellular wireless communication, dedicated short-range communication (DSRC), and IEEE802.11p standard. Vehicles collect information about their environment and neighboring vehicles via GPS or sensors, then broadcast relevant information. It is essential to guarantee that information arrives accurately and promptly at the correct place when sharing across the IoV network.

In the wireless communication mode, the communication network becomes more open. Malicious attackers may change the information transmitted on the IoV and pretend to be legitimate nodes to send false information through the IoV network. This may result in shared information being vulnerable to possible attacks [17], such as

replay attacks, masquerade attacks, man-in-the-middle attacks, and deny-of-service attacks. On the other hand, during secure message interactions, malicious attackers can deduce sensitive information through trajectory tracking, such as the user's identity and occupation. They may launch attacks that cause accidents, injuries, or even deaths [18].

The reason for the above is that we added new communication protocols for the vehicle, which further created many new vulnerabilities for hackers. Therefore, as long as there is a need to exchange information over IoV, the process of mutual identity authentication is essential, which is why information security and authentication have been an important research issue from the past to the present. To defend against multiple malicious attacks, designing a safe authentication scheme is the solution. At the same time, the Internet of Things (IoT) has emerged, allowing us to connect everything to the Internet and become increasingly dependent on related facilities. The burden on 5G mobile network-related devices increases rapidly with the number of IoT devices. Because of personal privacy, the service provider cannot store the user's data for improper purposes without the user's consent and should take necessary measures to hide the user's true identity. Therefore, this authentication scheme must also consider minimizing overhead and maintaining personal privacy.

On the other hand, as the vehicle usually moves frequently at high speed, it also leads to more frequent hand changes. J. Cao [19] and A. Sharma [20] have discovered that during the handover authentication process, there existed vulnerabilities of security, such as vulnerabilities against mutual authentication, key forward secrecy (KFS), denial of service (DoS) attacks, etc. Moreover, the current standard does not protect the user's private identity.

Following the traditional security authentication scheme, each authentication needs to rely on the participation of some registered devices. When many vehicles perform the authentication process simultaneously, many signaling messages may be generated, resulting in signaling congestion, which may block essential nodes. In addition, these transmitted signaling messages sometimes have the same content. However, considering intermediary attacks to tamper with messages, it is inevitable to use one-to-one single-point transmission to each other to avoid attacks by carrying the other party's unique identification on the message packets. To prevent the above signaling congestion and data transmission congestion problems, it is necessary to think of solutions that satisfy security and efficiency. Among them, the one-to-many transmission service group broadcast service can avoid repeated transmission of the same data on the same link, thus effectively reducing the consumption of network resources [21]. To satisfy the security and efficiency requirements of VANET, the current research direction of

VANET is towards group communication [22–24]. In recent years, the technology of mobile networks has matured, and there is more and more research on the combination of 5G and VANET. In the study of V2X, many schemes based on group correlation were also designed to satisfy the requirements of safety and high efficiency [25–27]. Therefore, grouping is a viable solution to meet the requirements of security and efficiency at the same time.

1.2 Motivation

This study addresses security, privacy protection, and computational and communication overhead in the batch verification process within the 5G-VANET environment. Currently, most vehicle authentication schemes rely on individual authentication mechanisms, leading to a significant increase in authentication delay as the number of vehicles grows, adversely affecting real-time communication performance in VANETs. Moreover, existing handover authentication schemes require multiple interactions with the 5G core network during verification, increasing computational and communication overhead. This may even place an excessive burden on 5G infrastructure, resulting in significant delays. At the same time, privacy protection remains a challenge in existing schemes. Some approaches fail to effectively prevent identity linkage attacks, allowing attackers to track vehicle identifiers and infer drivers' behavioral patterns and private data.

We proposed a scheme that can be split into the authentication and key agreement parts. In the authentication part, we combine the unidirectional hash function and ECC to design a lightweight and efficient algorithm used in identity authentication. The algorithm can be used to authenticate a batch of vehicle users in the coverage area at the same time. It can also be used in the handover phase to reduce its overhead cost without repeated authentication through the 5G core as much as possible. In the key agreement part, we design a tree-based key agreement algorithm and consider the scenarios of vehicles joining and leaving the group. When these two scenarios are triggered, the key is updated, which can help legitimize the vehicle to update the session key correctly. An elliptic curve signature protects the key update message. It does not contain any identifying information, which always ensures the identity privacy of the vehicle.

1.3 Contribution

Our main contributions are summarized as follows:

- (1) We proposed a lightweight cryptography technique consisting of elliptic cryptography and hash functions to design a Batch Authentication and Tree-based Key

Agreement scheme, abbreviated as BATKA, making the computational and communication overheads associated with authentication relatively low.

- (2) The batch authentication feature for VANETs in a 5G network architecture enables batch message authentication to reduce computational overhead, communication overhead, and storage overhead and to reduce authentication delays in VANETs. Allowing the receiver to perform batch authentication of multiple incoming messages is more efficient than a single authentication method.
- (3) We apply the idea of a group leader to VANET and modify the handover process by abandoning a group session key to authenticate handover and changing the key structure to a tree structure. Make it more flexible for new group members to join or leave.
- (4) We provided privacy protection properties for semi-trusted hardware providers, such as anonymity and unlinkability. Semi-trusted vendors cannot associate messages with specific vehicles, and no adversary can break the anonymity under security assumptions.
- (5) We conducted a security proof using a random oracle model under the Computational Diffie-Hellman Problem (CDHP) and the Elliptic Curve Discrete Logarithm Problem (ECDLP) assumptions to show that our proposed scheme satisfies the security. The ECDLP is a particular case of the discrete logarithm problem in which the cyclic group G is represented by the group $\langle P \rangle$ of points on an elliptic curve. It is of cryptographic interest because its apparent intractability is the basis for the security of ECC [28].
- (6) Maintain data integrity to avoid message modification during transmission. It can also be used in the key agreement stage to prevent man-in-the-middle attacks.

The structure of the rest of this paper is as follows: Section 2 introduces the related works referenced in our approach. Section 3 presents the proposed batch authentication algorithm and details the application process to a 5G system for authentication. Section 4 analyzes the security of our approach, first proving the unforgeability of the authentication algorithm approach against several adversaries and then analyzing the reliability of the proposed VANET attack and the security attributes achieved. Section 5 conducts a comparative analysis with other approaches. Section 6 concludes the paper and suggests directions for further improvement in the future.

Table 1 Acronyms and terms

RSU	The roadside unit
IoV	Internet of Vehicles
VANET	Vehicular ad hoc networks
C-V2X	Cellular-based Vehicle-to-Everything
DSRC	Dedicated Short-Range Communication
IoT	Internet of Things
KFS	Key Forward Secrecy
DoS	Denial of Service
IoE	Internet of Everything
PKI	Public Key Infrastructure
IBS	Identity-based Signatures
IBC	identity-based cryptography
RTA	regional trust authority
AMAC	Aggregated Message Authentication Codes
ECC	Elliptic Curve Cryptography
BP	bilinear pairings
CDHP	Computational Diffie-Hellman Problem
ECDLP	Elliptic Curve Discrete Logarithm problem
MME	Mobility Management Entity
TPD	tamper-proof device
gNB	Next Generation Node B
SQN	Sequence Number
AV	Authentication Vector
GLV	the group leader
PID	pseudo-identity
ROM	Random Oracle Model
PPT	probabilistic polynomial time
PBC	Pairing-Based Cryptography library
pypbc	python-based pairing-based cryptography

2 Related works

This section presents the relevant developments in technology and design referenced in this paper. The acronyms and terms used in this paper are listed in Table 1.

2.1 Group-based batch authentication schemes

With the advancement of the IoE, 5G infrastructure has become increasingly mature. The development of vehicular networks has led to the establishment of more regulations and standards. At the same time, related research has continuously improved based on the efforts of previous studies. Verification delays can lead to severe consequences [29]. Verification efficiency depends on computational costs and communication overhead, where lower computational costs ensure faster vehicular communication [30]. Additionally, as the number of vehicles increases, multiple messages must be verified simultaneously, requiring solutions to manage the resulting signaling and computational overhead.

Considering the aforementioned security concerns, numerous studies have proposed various measures. For instance, digital signatures have been used to verify message authenticity, and anonymous message authentication protocols incorporating pseudonym mechanisms have been introduced to achieve conditional privacy protection [31, 32]. Later, implementations based on third-party Public Key Infrastructure (PKI) emerged, allowing users to load public-private key pairs and certificates from a trusted third party [33, 34]. Rahbari et al. proposed a cryptographic approach leveraging PKI infrastructure to detect impersonation attacks, wherein an attacker node simulates multiple nodes within the network to deceive others and extract sensitive information. To prevent impersonation attacks, users must verify certificate validity before using a public key, leading to significant certificate management overhead. To address this issue, identity-based signatures (IBS) were introduced to achieve conditional privacy protection in VANETs [35]. Subsequently, more advanced identity-based cryptography (IBC) protocols were developed, deriving public keys from identifiers and extending PKI-based protocols [36, 37]. One common limitation of these protocols is requiring vehicles to change pseudonyms periodically to maintain privacy and information security. This results in high computational and storage costs associated with generating new pseudonyms and verifying identity legitimacy. To mitigate these overheads, researchers have explored various alternative methods.

Group signature [38] is a specialized digital signature technique that allows group members to sign a shared message while keeping their identities hidden from the public, except for a designated authority (e.g., a manager) who can reveal the signer. This mechanism prevents group members from abusing their anonymity while maintaining privacy. Harn et al. [39] introduced the concept of group verification, aiming to reduce network burden and resource wastage, and applied it to 5G-based NB-IoT scenarios. As the name suggests, group verification seeks to authenticate multiple devices through a single verification process.

Lee and Lai [40] proposed a batch-verifiable authentication scheme. Still, it was later found that their approach failed to ensure user privacy, as the sender's true identity could be obtained by other vehicles or third parties [41]. Additionally, their scheme could not resist forgery attacks and did not satisfy non-repudiation requirements, making it impossible to trace unauthorized entities and leaving room for malicious users to exploit. Recently, Tzeng et al. [42] proposed an identity-based enhanced batch verification scheme for V2I and V2V communication in VANETs, demonstrating that Lee and Lai's scheme was vulnerable to identity privacy breaches, forgery, and anti-tracking attacks.

A group key establishment mechanism is necessary for secure and reliable group communication with

data-sharing capabilities. Kim et al. [43] introduced a tree-based group Diffie-Hellman key agreement protocol, supporting node addition, removal, group merging, and splitting. Their approach offers significant efficiency improvements compared to traditional group key agreement protocols. However, despite its proven security and efficiency, one of its requirements makes it unsuitable for key management in V2 V scenarios. Yanji Jiang et al. proposed enhancing VANET security by incorporating a regional trust authority (RTA) [44].

Ring signatures provide an alternative to group signatures, where a signer can construct a ring including their public key and those of other group members, forming a circular structure. The signer then generates a ring signature using their private key and the public keys of others. The signature can be verified to confirm that a group member signed it, but the actual signer's identity remains anonymous. Essentially, this is a group signature without a central manager.

To further enhance authentication efficiency, some studies have combined pseudonym certificates with batch processing [45, 46]. Early research by Y. Liu [46] introduced a batch authentication system, which authenticates multiple messages simultaneously using a proxy vehicle for message verification. The RSU then verifies the proxy vehicle's output to ensure message integrity, authenticity, and non-repudiation. More recent studies have focused on RSU-based batch verification for authenticating multiple vehicles within its coverage area simultaneously [47–50]. These findings confirm that batch verification can significantly reduce authentication overhead, improving computational efficiency in VANET environments. However, these methods have several limitations; first, signature and verification operations are highly time-consuming, causing delays in vehicle decision-making, which increases risks in autonomous driving within VANETs: high storage and communication overheads for vehicle users. Significant

storage and bandwidth are needed since different pseudonym certificates are required for each authentication process. Most schemes assume RSUs are trustworthy, but RSUs are typically deployed by different organizations. They may be only semi-trusted, potentially leading to privacy breaches.

Despite extensive research on VANET authentication in 5G environments, many existing studies fail to focus on authentication and handover mechanisms for 5G-V2X vehicles. Existing solutions are often weak in security and efficiency. G. Li et al. [51] proposed a queue authentication handover protocol for 5G-V2X networks using Aggregated Message Authentication Codes (AMAC) to reduce signaling overhead. However, the protocol requires multiple-point multiplication operations, making it highly time-consuming. Moreover, it fails to protect vehicle privacy. It is vulnerable to DDoS attacks since identity packets are transmitted to group leaders in plaintext through insecure channels. Attackers can disrupt the entire group's authentication process by continuously sending fake MACs to the group leader. This issue exists in all aggregation-based protocols, with some attempts to mitigate it using Divide-and-Conquer algorithms—but at the cost of additional computational overhead. Finally, Table 2 A review of authentication schemes, summarizes the authentication research and related methodologies discussed above.

To address these challenges, [27] proposed a lightweight cryptographic approach for efficient vehicle authentication and re-authentication in 5G networks, reducing computational overhead. However, it incurs high communication costs and does not protect vehicle privacy. X. Yan and L. Wei proposed unique group verification methods. They employed a tree structure to facilitate member updates and related information management within the group [52, 53].

Table 2 A review of authentication schemes

Scheme	Public Key Infrastructure	Signature/Cryptography	Batch authentication
A. Wasef et al. [33]	Yes	N/A	No
J. Kang et al. [34]	Yes	N/A	No
C. Zhang et al. [35]	Yes	Identity-based signature	No
N. W. Lo et al. [36]	Yes	Identity-based Cryptography	No
J. Cui et al. [37]	Yes	Identity-based Cryptography	No
C. Delerablée et al. [38]	No	Group signature	No
Jiang et al. [44]	No	Group signature	No
A. Wasef et al. [45]	No	N/A	Yes
Y. Liu et al. [46]	No	N/A	Yes
G. Li et al. [51]	No	N/A	Yes

2.2 Feature comparison of authentication schemes

Many researchers have spent considerable effort developing identity-based authentication schemes to improve efficiency. These schemes utilize IBS without the use of certificates for identity authentication. Identity-based authentication in these schemes uses pseudonyms for communication over the IoV network to protect the privacy of user identities. These schemes can be broadly classified into two categories: one is an authentication scheme based on bilinear pairing operations, and the other is based on elliptic curve authentication schemes. ECC-based designs do not require bilinear pairing.

In 2016, an anonymous batch authentication scheme known as ABAH was proposed by Jiang et al. [54], which uses bilinear pairing and performs the Certificate revocation list-checking process by computing the hash message authentication code. However, [54] lacks an efficiency scheme because the group key agreement requires regular updates and revocation of group keys and the high overhead caused by bilinear pairing when handling batch authentication. In 2018, Ahamed et al. proposed a new identity authentication scheme [55]. The scheme allows anonymous mutual authentication between vehicle users, and a set of batch keys is designed for batch authentication through each vehicle to reduce the computational overhead of identity verification. An effective certificateless aggregate signature scheme was proposed by Q. Mei et al. in 2021, using aggregation technology to reduce computational overhead and resources of the bandwidth [56]. However, there is a potential risk of tampering attacks by later discovery and huge computational overheads and communication overheads because of the use of bilinear pairing and mapping to point hash operations.

All the above studies have a common flaw of high overhead due to bilinear pairing. Cryptographic research has indicated that bilinear pairing is the most expensive of all cryptographic operations in terms of computation and

storage. On the other hand, ECC has a shorter key length than RSA. When the ECC with the 160-bit key has the same level of security as the RSA with the 1,024-bit key [57], it can effectively improve computing efficiency, communication efficiency, and storage capacity. In 2017, J. Cui et al. proposed a scheme called SPACF [58], which aims to solve the problem of existing solutions relying heavily on tamper-proof devices (TPD) and their inability to satisfy security requirements. During the batch authentication phase, [58] introduces the binary search method and Cuckoo filter to enhance the performance. However, due to the frequent communication interactions, this scheme is relatively inefficient in practical applications. In 2020, Anil Kumar Sutrala et al. proposed a new authentication mechanism in the IoV environment using the ECC and conditional privacy protection batch-authentication methods [59]. This mechanism allows a vehicle to authenticate its neighboring vehicles. At the same time, the RSU can also authenticate nearby vehicles in batches, reducing the time required to authenticate individual messages. However, it was discovered that there was a risk of being forged and attacked the following year [60].

Mahmood A. Al-Shareeda et al. proposed in 2022 a chaotic map-based conditional privacy-preserving authentication scheme, which employs Chebyshev polynomial mapping operations and a chaotic map-based hash function to enhance security in 5G-supported vehicular networks while eliminating the need for BP and elliptic curve computations (ECC computations) [61]. Badiea Abdulkareem Mohammed et al. developed in 2023 ANAA-Fog, an anonymous authentication scheme for 5G vehicular fog computing. This mechanism allows fog servers to generate temporary private keys for participating vehicles, facilitating digital signature verification and reducing computational and communication overhead [62]. A. Almazroi et al. proposed in 2023 an efficient certificateless authentication scheme, supporting pseudonyms and traceability, designed explicitly for 5G-assisted

Table 3 Comparison of authentication features

	Jiang et al., 2016 [54]	Ahamed et al., 2018 [55]	Q. Mei et al., 2021 [56]	J. Cui et al., 2017 [58]	Anil Kumar Sutrala, 2021 [59]	Almazroi et al., 2023 [61]	Al-Shareeda et al., 2022 [62]	Mohammed et al., 2023 [63]
Bilinear Map	Yes	Yes	Yes	No	No	No	No	No
Pseudonymous authentication	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Special design	Certificate revocation list	Batch key	Certificateless	Cuckoo filter & Binary Search	N/A	Certificateless	Chaotic-Map-ping-based	Certificateless
Computational overhead	High	High	High	Low	Low	Low	Low	Low
Information Security Risk	N/A	N/A	Yes	Yes	Yes	No	No	No

vehicular fog computing. Their approach utilizes fog servers supported by 5G base stations to perform ECC operations, enhancing authentication security and efficiency through partial pseudonym ID and private key distribution mechanisms [63].

Finally, we summarize the contents of the above-mentioned authentication schemes for IoV in Table 3. Table 3 shows that there are still some problems with the safety and efficiency of the authentication scheme for IoV. To solve these problems, we propose a secure and efficient ECC-based batch authentication scheme for privacy protection without pairing conditions. BATKA satisfies the security requirements of anonymous authentication, unlinkability, traceability, and message integrity. BATKA has low computational, communication, and storage costs.

3 System model and proposed scheme

3.1 Preliminaries

This subsection presents the assumptions that need to be satisfied by the method proposed in this section and an introduction to the system model.

3.1.1 Computational diffie-hellman problem (CDHP)

CDHP is a related problem of the Diffie-Hellman key exchange protocol, which describes the difficulty of computing the results of a specific Diffie-Hellman calculation when the Diffie-Hellman parameters are known. Given an elliptic curve E , suppose there is an additive cycle group G on E with P as the generator and order q . And $G \in Z_q^*$ is given two parameters $a, b \in Z_q^*$, so that $aP \in G$ and $bP \in G$ are satisfied, and the CDHP is to find the solution of abP [64, 65]. Computational Diffie-Hellman assumption means that there is no such a probabilistic polynomial time Turing machine.

3.1.2 Elliptic curve discrete logarithm problem (ECDLP)

Let two huge prime numbers, p and q , and a finite field F_p composed of Np elements. There is an elliptic curve defined as $y^2 = x^3 + ax + b(modp)$, where $a, b \in F_p$ must satisfy $4a^3 + 27b^2(modp) \neq 0$. Suppose now there is an infinitely far point Θ , the point Θ and all other points are additive groups G with order q from the elliptic curve, and P is the generator of this group G . And there is a scalar multiplication group defined as $kP = P + P + \dots + P$, where $k \in Z_q^*$. Given two points $P, Q \in G$ on an elliptic curve, where $Q = xP$ and $x \in Z_q^*$, the ECDLP is to find an integer k such that $kP = Q$. This problem is a discrete logarithm problem, and it is still difficult to calculate the value of k . Discrete Logarithm

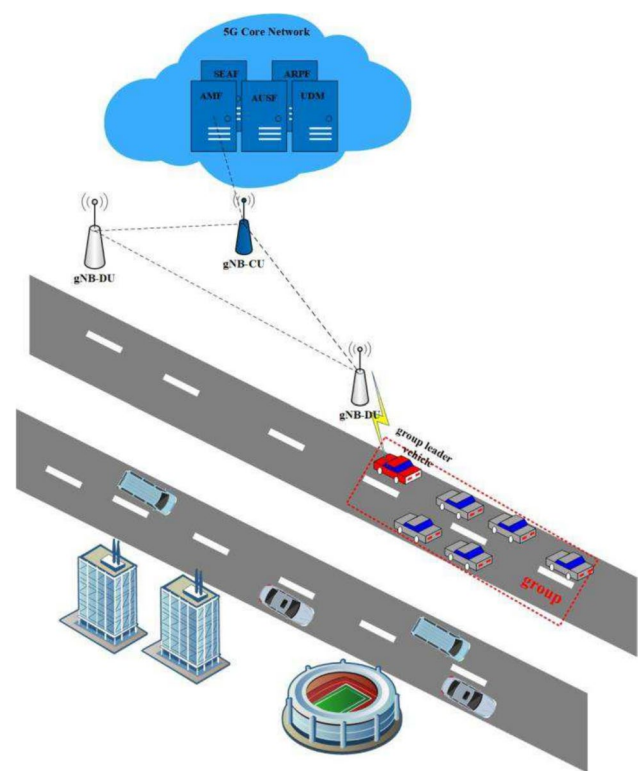


Fig. 1 System model

assumption means no such probabilistic polynomial time Turing machine exists.

3.2 System model

This subsection introduces our system model, as shown in Fig. 1, divided into two main parts: VANET and 5G core network.

3.2.1 5G core network model

This subsection introduces our system model's 5G core network components. First, we introduce the basic functions and special components from the security perspective because they are used for identity authentication.

- AMF: Access and mobility function

AMF plays a similar role in 5G networks as a Mobility Management Entity (MME) in 4G. It is responsible for managing access control, mobility management, authentication, authorization of user devices, and some UE-related information. AMF ensures that users have smooth access to the 5G network and stay connected while on the move while

handling the important tasks of registration, authentication, and authorization.

- SEAF: Security anchor function

SEAF is an AMF subcomponent and a "middleman" in the 5G network. It can deny UE authentication but needs to rely on UE's local network for the authentication process to ensure the network's security and prevent unauthorized access. It is also used to generate a key, K_{SEAF} for subsequent communication between the user and the serving network.

- AUSF: Authentication server function

The 5G network has functions similar to MME and HSS. It mainly supports user identity authentication, receives authentication requests from SEAF, and interacts with ARPF to complete the subsequent authentication and challenge-response process. This ensures that users and devices in the 5G network obtain legitimate access and provide communications security.

- UDM: Unified data management

UDM is to provide a unified data management mechanism to ensure data security and effective use, including ARPF components. It supports authentication, access authorization, and user data management to ensure secure and legitimate access for subscribers in the 5G network, process subscription information, generate authentication credentials, and manage user identifiers.

- ARPF: Authentication credential repository and processing function

ARPF is a critical component of the 5G network responsible for storing, processing, and managing the certificate and related parameters required for authentication. It assumes the key role of generating authentication vectors (AVs) and cryptographic method processing to ensure the security and validity of the authentication process. At the same time, it implements stringent security measures to protect the confidentiality and integrity of the stored sensitive data. It interacts with other 5G components to ensure users are granted legitimate access.

3.2.2 VANET model

This subsection focuses on the VANET components used in our system model.

- OBU: On-Board Unit

OBU is an important device installed on vehicles that enables real-time communication and data exchange with other vehicles, infrastructure, and cloud services through wireless communication, positioning, and data collection functions. It provides functions such as vehicle navigation, traffic condition monitoring, vehicle data collection and analysis, safety alerts, and vehicle management, greatly enhancing vehicle intelligence, safety, and efficiency. It is also a more convenient and efficient operation management tool for vehicle owners and fleet managers.

- Vehicles:

Each vehicle has an OBU that generates messages, transmits them to nearby vehicles and APs, and stores sensitive information in its TPD.

3.3 Proposed scheme

This subsection introduces our proposed approach for group batch authentication. The system architecture is mainly modified regarding [52, 66]. [66] combines the existing method of aggregated signatures with the group leader's idea to effectively reduce the overhead required for authentication. [52] uses a group of group session keys as the handover scenario of the 5G-V2X application to reduce the overhead during handover. Therefore, we apply the idea of a group leader in [66] to VANET and modify the handover process in [52] by abandoning a group session key to authenticate handover and changing the key structure to a tree structure. Make it more flexible for new group members to join.

3.3.1 System initialization

Next is the initial system phase. The system uses the following steps to generate the initial system parameters and various parameters required by the public and private key pairs. This part is mainly performed by the AMF and SEAF of the 5G core network. The steps for this phase are as follows:

- **Step 1:** Given a safety factor k , AMF can determine a large prime number $q (q > 2^k)$.
- **Step 2:** AMF selects P as the generator of G to generate an elliptic curve additive group G with order q .
- **Step 3:** AMF randomly selects a number, $K_{mas} \in Z_q^*$ as the master key and then calculates its corresponding public key $P_{pub}, P_{pub} = K_{mas}P$.
- **Step 4:** Generate the key pair of the system. SEAF randomly selects a number $K_{pri} \in Z_q^*$ as the master private key and then calculates the master public key $K_{pub}, K_{pub} = K_{pri}P$ as the primary public key of the system.

- **Step 5:** Generate four secure hash functions:
 $H_1 : G \rightarrow Z_q^*$, $H_2 : \{0, 1\}^* \rightarrow Z_q^*$, $H_3 : \{0, 1\}^* \rightarrow Z_q^*$,
 $H_4 : \{0, 1\}^* \rightarrow Z_q^*$
- **Step 6:** AMF announces release system parameters
 $(G, q, P, K_{pub}, P_{pub}, H_1, H_2, H_3, H_4)$

3.3.2 Group leader vehicle registration

When a vehicle broadcasts through VANET that there is no group leader vehicle in the vicinity, a vehicle is registered as a group leader and performs the following registration steps:

- **Step 1:** When vehicle V is activated, a number $x_i \in Z_q^*$ is randomly selected as the ECC partial private key and then $X_i = x_i P$ is calculated as the ECC partial public key.
- **Step 2:** Vehicle V sends Initial_Group_Leader_Registration Request with 5G-GUTI or SUCI and the partial public key, X_i . We generated in step 1 to AMF. We use the Elliptic Curve Integrated Encryption Scheme (ECIES) to encrypt SUPI into SUCI.
- **Step 3:** After receiving the Initial_Group_Leader_Registration Request, AMF sends a Nausf_UE_Authentication_Authenticate Request message to SEAF. SEAF then forwards to AUSF, which contains $(X_i, SUCI)$.
- **Step 4:** AUSF first uses the subscription identifier de-concealing function to decrypt SUCI into SUPI and then sends Nudm_UE_Authentication_Get Request to UDM.
- **Step 5:** UDM/ARPF selects a random number RAND to generate a Sequence Number (SQN) and generates an Authentication Vector (AV) consisting of AUTN, XRES, K_{AUSF} , and five one-way hash functions (f1, f2, ..., f5). The 5G-AKA process is followed here. Then, the authentication response (Nudm_Authentication_Get Response), composed of AV, is transmitted to AUSF and SUPI. Table 4 shows the functionality of the five one-way hash functions.

$$\begin{aligned} MAC &= f1(RAND)_K \\ XRES &= f2(RAND)_K \\ CK &= f3(RAND)_K \\ IK &= f4(RAND)_K \\ AK &= f5(RAND)_K \\ AUTN &= (SQN \oplus AK || AMF || MAC) \\ K_{AUSF} &= KDF(CK, IK) \end{aligned}$$

- **Step 6:** AUSF performs authentication according to UDM Nudm_Authentication_Get Response. If the authentication fails and K_{AUSF} If it is discarded, the process is terminated. Otherwise, store K_{AUSF} and generate H_{XRES} .
 $H_{XRES} = h(XRES)$
- **Step 7:** AUSF sends Nausf_UE_Authentication_Authenticate Response to SEAF, which contains RAND, AUTN, and K_{XRES} .
- **Step 8:** SEAF stores H_{XRES} , and forwards RAND and AUTN to AMF.
- **Step 9:** After the AMF receives the UE_Authentication_Authenticate Response from the AUSF, the system parameters mentioned in the initial phase of the system are returned to the UE.
- **Step 10:** UE calculates the RES, returns the Authentication Response to SEAF, and continues the 5G-AKA process.
- **Step 11:** Through the system parameters obtained in Step 9, the real identity RID_i is calculated $\alpha_i = H_1(RID_i)$ and transferred to AMF, which contains the user's accurate information, such as the owner's ID card number, license plate number, and vehicle identification number.
- **Step 12:** AMF calculates $TID_1 = H_1(SUCI || K_{AMF})$ and $GID_i = H_2(H_1(SUCI || K_{AMF}) + \alpha_i)$ and stores this GID_i and TID_1 , where temporary id (TID) and group id (GID).

Table 4 Functionality of the five one-way hash functions

One-way hash function	Functionality
f1 (RAND) _K = MAC (Message Authentication Code)	Computing a MAC ensures integrity verification, ensuring message integrity, and authentication
f2 (RAND) _K = XRES (eXpected RESponse)	Compute XRES as the authentication response When the UE calculates and returns RES (Response), the 5G network compares XRES and RES. If they match, the verification is successful
f3 (RAND) _K = CK (Cipher Key)	Derive the Cipher key to protect the confidentiality of data transfers
f4 (RAND) _K = IK (Integrity Key)	Compute the Integrity key to protect the integrity and prevent tampering
f5 (RAND) _K = AK (Anonymity Key)	Generate the Anonymity Key (AK) to conceal the Sequence Number (SQN) and prevent replay attacks

- **Step 13:** AMF generates its pseudo identity (PID) for the vehicle V_{GL} which is the group leader. SEAF randomly selects $\mu_i \in Z_q^*$ and calculates $PID_{v_{GL}1} = \mu_i P$ and $PID_{v_{GL}2} = a_{GLi} \oplus H_2(\mu_i P_{pub} || T_{v_i})$. Let $PID_{v_{GL}} = \{PID_{v_{GL}1}, PID_{v_{GL}2}, T_{v_{GL}}\}$ of V_i , where $T_{v_{GL}}$ is the valid period of $PID_{v_{GL}}$.
- **Step 14:** Then SEAF calculates the partial private key of V_i $pmk_{v_{GL}} = U_i K_{pri} + \mu_i$, where $U_i = H_3(PID_{v_{GL}} || K_{pub})$, and sends the result to AMF.
- **Step 15:** AMF sends the Initial_Group_leader_Registration Request to the group leader according to GID, which contains $\{GID_i, TID_1, pmk_{v_{GL}}, PID_{v_{GL}}\}$.
- **Step 16:** The group leader randomly selects $r_i \in Z_q^*$ as its private key and calculates $S_i = r_i P$ and calculates $h_i = H_4(M_i || PID_{v_{GL}} || T_i)$ and $C_i = H_3(PID_{v_i} || GID)$, where T_i is the current timestamp. And $\phi_i = pmk_{v_i} + r_i h_i + r_i C_i$, used to generate the group leader's signature $\sigma_i = \{\phi_i, S_i\}$. Complete this registration.

3.3.3 Vehicle registration

The vehicle generates unique authentication parameters for subsequent identity authentication, as shown in Fig. 2.

- **Steps 1–3:** Vehicle V_i initiates an Initial_Join_Group Request. The Group leader calculates $TID_2 = H_1(RID_i)$ and return parameters $\{GID, TID_1, TID_2, K_{pub}, H_1\}$ to the vehicle V_i .
- **Step 4:** When receiving the message from the group leader (GLV), the vehicle V_i first checks whether the equation of $GID = H_2(TID_1 + TID_2)$ is true. If it is not an equation, it has encountered a fake group leader and stops the subsequent process.
- **Step 5:** Vehicle V_i uses the K_{pub} and H_1 received from the group leader to calculate $\alpha_i = H_1(RID_i)$ and sends an Initial Group Registration Request to AMF, which contains $\{GID, SUCI, \alpha_i\}$ and sends it to AMF, Vehicle V_i will also be sent α_i to the group leader.
- **Steps 6–7:** AMF receives the V_i Initial_Group_Registration Request from the vehicle. It first checks whether the GID exists. If it exists, continue the f process, and AMF forwards it to AUSF.
- **Steps 8–9:** AUSF first uses the subscription identifier de-concealing function to decrypt SUCI into SUPI and then sends Nudm_Group_Member_Authentication_Get Request to UDM.
- **Steps 10–12:** AUSF performs authentication according to UDM_Nudm_Group_Member_Authentication_Get Response. If the authentication fails, terminate the process. On the contrary, AUSF sends the initial group registration response to SEAF.

- **Steps 13–14:** SEAF generates its pseudo-identity (PID) for V_i . SEAF randomly selects $\mu_i \in Z_q^*$ and calculates $PID_{v_i1} = \mu_i P$ and $PID_{v_i2} = a_i \oplus H_2(\mu_i P_{pub} || T_{v_i})$. Let V_i of $PID_{v_i} = \{PID_{v_i1}, PID_{v_i2}, T_{v_i}\}$, where T_{v_i} is valid for PID_{v_i} the term. And record PID_{v_i} in the exclusive L_{GID} list under the GID.
- **Step 15:** Then SEAF calculates the partial private key of V_i $pmk_{v_i} = U_i K_{pri} + \mu_i$, where $U_i = H_3(PID_{v_i} || K_{pub})$, and transmits it to AMF.
- **Step 16:** AMF sends an Initial_Join_member Request to the group leader according to the GID, which contains $\{\alpha_i, pmk_{v_i}, PID_{v_i}\}$.
- **Step 17:** The group leader checks whether α_i has been stored or not. If not, it sends an updated GID request. Otherwise, it sends $\{pmk_{v_i}, PID_{v_i}\}$ to the corresponding V_i according to α_i . V_i will store $\{H_3, H_4, pmk_{v_i}, PID_{v_i}\}$ after receiving it.
- **Step 18–1:** V_i generates its signature. V_i randomly selects $r_i \in Z_q^*$ as its private key and calculates $S_i = r_i P$.
- **Step 18–2:** V_i calculates $h_i = H_4(M_i || PID_{v_i} || T_i)$ and $C_i = H_3(PID_{v_i} || GID)$, where T_i is the current timestamp. And $\phi_i = pmk_{v_i} + r_i h_i + r_i C_i$, used to generate a signature $\sigma_i = \{\phi_i, S_i\}$.
- **Step 18–3:** Broadcast the message signature to other members in the group $\{GID, PID_{v_i}, \sigma_i, T_i\}$.

3.3.4 Batch authentication

When the vehicle can broadcast or receive n messages at the same time $< \{GID, M_1, PID_1, \sigma_1, T_1\}, \{GID, M_2, PID_2, \sigma_2, T_2\}, \dots, \{GID, M_n, PID_n, \sigma_n, T_n\} >$, especially in dense traffic areas. If the recipient authenticates messages individually, this may result in redundant calculations and relatively long delays. Batch authentication is performed using the following steps:

- **Step 1:** The vehicle receiving the message first checks whether the GID is in the same group as the group to which it belongs.
- **Step 2:** The vehicle checks the validity of its timestamp $T'_i - T_i > \Delta T$, it means that it has expired, and the message is rejected, where T_i means checking the timestamp of receiving the message, and ΔT is the transmission time difference between the two.
- **Step 3:** A new list of signatures with valid PIDs $\{\sigma_i\}_{i=1, \dots, n'}$ is generated, and a random vector $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_{n'}\}$ is chosen, here $\lambda_i \in \{0, 1\}^l$, l is usually 80 in VANET [67].
- **Step 4:** The vehicle calculates $U_i = H_3(PID_{v_i} || K_{pub})$ and $h_i = H_4(M_i || PID_{v_i} || T_i)$ and $C_i = H_3(PID_{v_i} || GID)$, and check whether the following equations are valid. If it is valid, the authentication is passed. The authentication

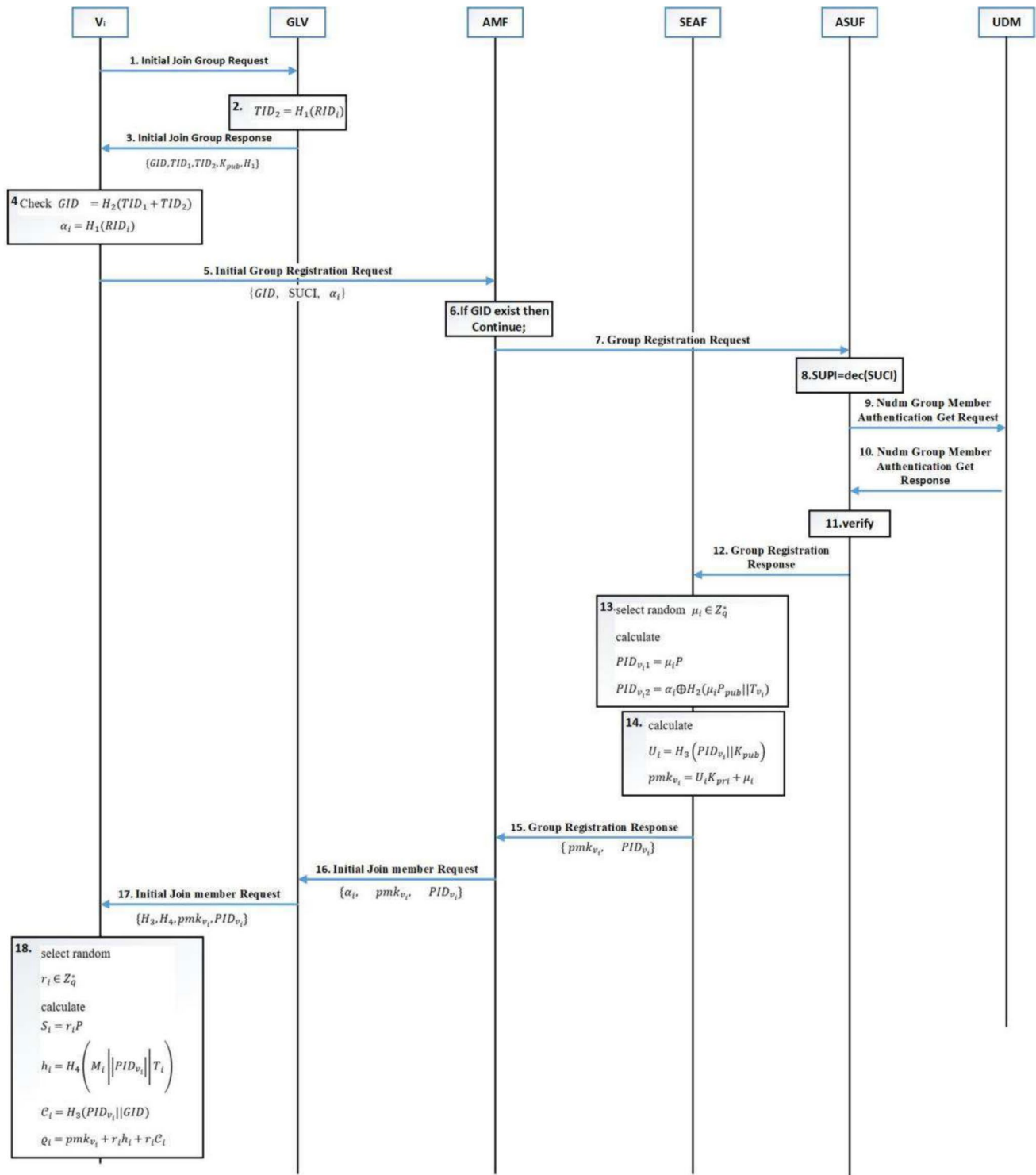


Fig. 2 Proposed initial registration phase

process ends, and then the Tree-Based Authenticated Key Agreement begins.

$$\left(\sum_{i=1}^{n'} \lambda_i Q_i \right) P = \left(\sum_{i=1}^{n'} \lambda_i U_i \right) K_{pub} + \sum_{i=1}^{n'} \lambda_i (PID_{i1} + (h_i + C_i) S_i)$$

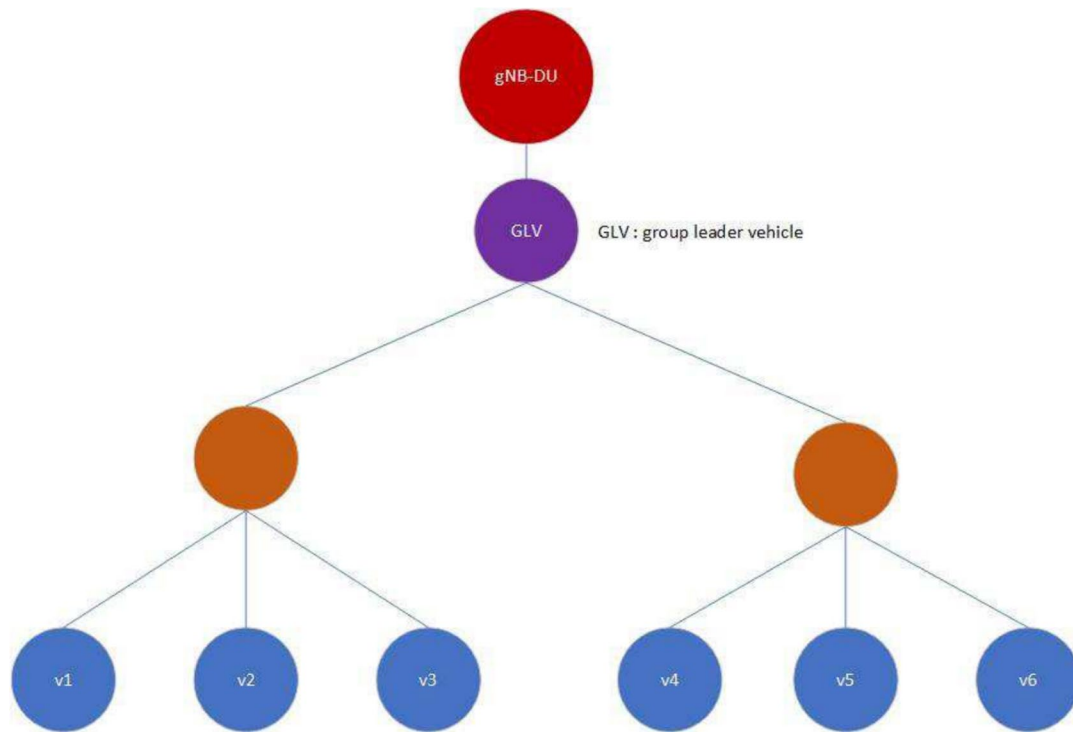


Fig. 3 Tree-based structure

3.3.5 Tree-based key agreement

Here, we refer to [53] to allow members a more flexible way to join a group or update members, as shown in Fig. 3.

- **Step 1:** Each vehicle has a safety factor k and can decide a prime number $q (q > 2^k)$ by itself. The vehicle randomly selects a number, $a_{vi} \in Z_q^*$ as v_i a private key and then calculate v_i public key $b_{vi}, b_{vi} = g^{a_{vi}} \bmod q$. g acts as a generator of a cyclic group G on an elliptic curve. Therefore, vehicles v_i each have their own private and public keys a_{vi} and b_{vi} . In the same way, the vehicle that is the group leader also has its own private key and public key a_{GL} and b_{GL} .
- **Step 2:** The vehicle broadcasts its GID and $\{b_{vi}, PID_i, \sigma_i, T_i\}$ to the group leader for authentication with our proposed authentication scheme.
- **Step 3:** After the group leader is authenticated, the group leader makes a group of three vehicles by broadcasting and agrees on a common key for this group. The group leader's vehicle r_{GL} and its group member vehicle V_i generates a random value r_{GL} and r_{vi} in each of them.
- **Step 4:** V_i and V_{i+1} will each generate new public keys c_{vi} and c_{vi+1} with their respective random values. Here are $c_{vi} = (b_{vi+1})^{a_{vi}r_{vi}} \bmod q$ and $c_{vi+1} = (b_{vi})^{a_{vi+1}r_{vi+1}} \bmod q$, respectively.

- **Step 5:** Both V_i and V_{i+1} calculate and generate a common private key and public key based on their respective r_{vi}, r_{vi+1} , and c_{vi}, c_{vi+1} received from the other party. The private key generated on the V_{vi} side is $K_{vi} = (c_{vi+1})^{r_{vi}} \bmod q$; on the V_{i+1} side $K_{vi+1} = (c_{vi})^{r_{vi+1}} \bmod q$.
- **Step 6:** When all the vehicles in the group have negotiated with V_{vi} , V_{vi} uses the common key material of the whole group to negotiate the key with the group leader, as well as the common public key and private key of the group and the group leader.
- **Step 7:** Finally, the group leader negotiates the key with the gNB to agree on a shared public and private key.

When another group joins, it performs batch authentication with the group leader. After authentication, it negotiates with the group leader to update the tree key according to the group's pre-negotiated common key, shown in Fig. 4.

There are two situations when leaving a group of vehicles, shown in Fig. 5. In the first case, the group leader notifies the remaining members to update their keys if they leave the group. The second case is to change groups. At this time, in addition to renewing the key in the first case, the members who are leaving are renegotiated and separated from

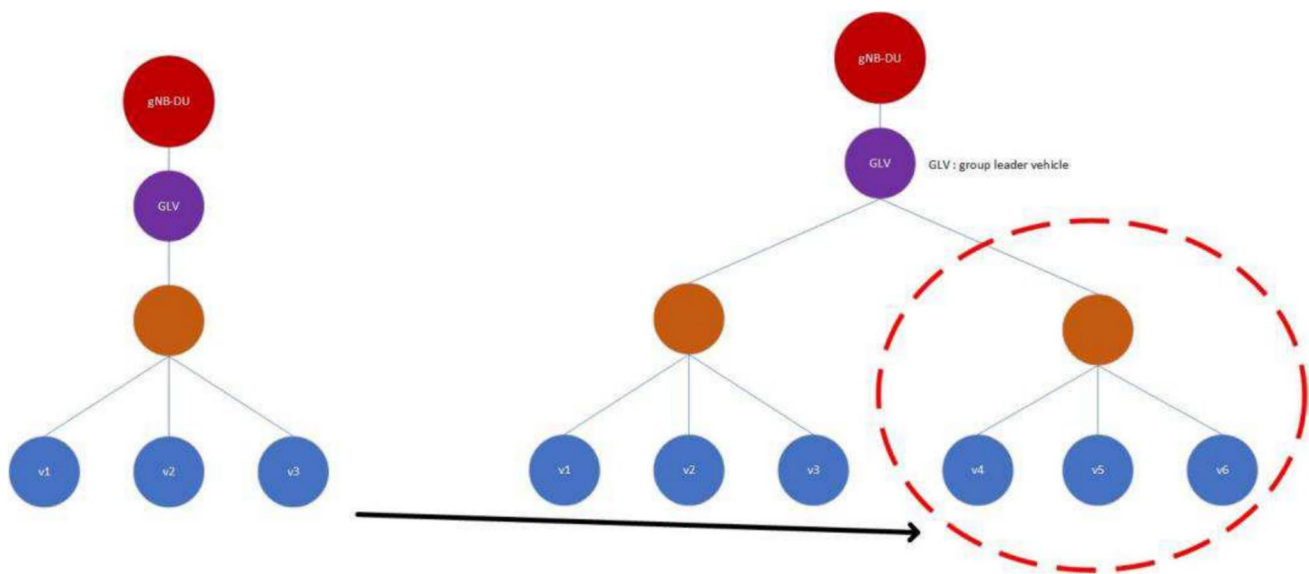


Fig. 4 A new vehicle joins

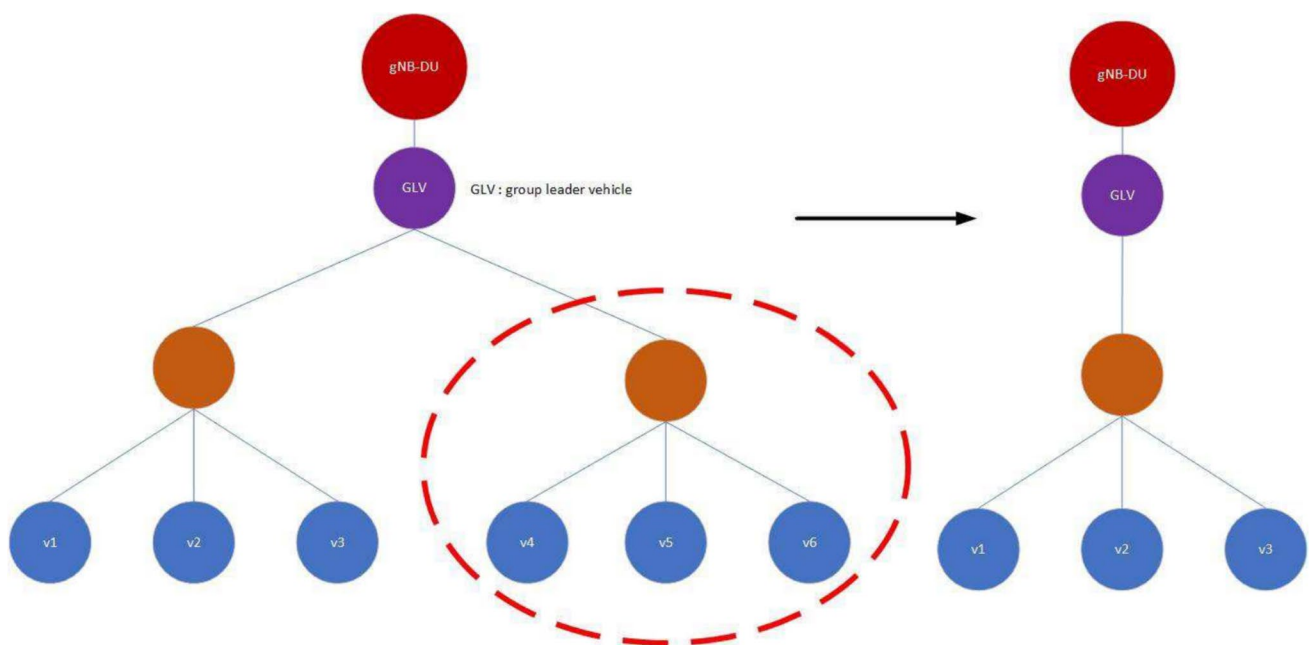


Fig. 5 A group of vehicles leave/remove

the original group to become a new group. After that, they can re-add the new group using the above joining method.

3.3.6 Group handover

When the group leader is leaving the original s-gNB and is about to enter the communication area of the t-gNB signal, that is, when the handover occurs, the following steps are executed.

- **Step 1:** The group leader initiates a group handover required to t-gNB, which contains $(GID, TID_1, \sigma_i, T_i)$.
- **Step 2:** After receiving the message, the t-gNB checks the validity of the timestamp T_i . If $T'_i - T_i > \Delta T$, t-gNB thinks that the message has expired for a long time and discards the message; otherwise, continue the following authentication process, where T_i represents the time stamp when the message was received, and ΔT is the difference in transmission time between them.
- **Step 3:** After receiving the message, t-gNB initiates Group_Authentication_data_Get to AMF, which contains GID_i and TID_1 .
- **Step 4:** AMF compares whether the GID is a legal group and whether it is the corresponding TID_1 . And return the corresponding system parameters $\{PID_{v_{GL}}, K_{pub}, H_3, H_4\}$.
- **Step 5:** t-gNB calculates $U_i = H_3(PID_{v_{GL}} || K_{pub})$ and $h_i = H_4(PID_{v_{GL}} || T_i)$. If the following equation holds, then the authentication is passed.

$$\rho_i P = U_i K_{pub} + PID_{v_{GL}1} + h_i S_i + C_i S_i$$
- **Step 6:** t-gNB requests group key data download from AMF.
- **Step 7:** AMF then sends the key transfer request with s-gNB.
- **Step 8:** The s-gNB forwards data to the t-gNB via the gNB-CU.

4 Security analysis

In this section, we discuss the security of the authentication process for the 5G-VANET scheme proposed in Section 3. First, we use the random oracle model that should be used for security analysis to prove its security, and then analyze some common security threats to confirm that the proposal achieved the security goal.

4.1 Correctness proof

In this subsection, we prove the correctness of the formulas used for batch authentication and handover authentication in subsections 3.3.4 and 3.3.6.

4.1.1 Correctness proof of batch signature authentication

The correctness of the batch authentication equation is as follows:

$$\left(\sum_{i=1}^{n'} \lambda_i \rho_i \right) P = \sum_{i=1}^{n'} \lambda_i (pmk_{v_i} + r_i h_i + r_i C_i) P$$

$$\begin{aligned} &= \sum_{i=1}^{n'} \lambda_i (U_i K_{pri} + \mu_i + r_i h_i + r_i C_i) P \\ &= \left(\sum_{i=1}^{n'} \lambda_i U_i \right) K_{pub} + \sum_{i=1}^{n'} \lambda_i PID_{i1} + \sum_{i=1}^{n'} \lambda_i h_i S_i + \sum_{i=1}^{n'} \lambda_i C_i S_i \\ &= \left(\sum_{i=1}^{n'} \lambda_i U_i \right) K_{pub} + \sum_{i=1}^{n'} \lambda_i (PID_{i1} + (h_i + C_i) S_i) \end{aligned}$$

4.1.2 Correctness proof of handover authentication

The correctness of the signature authentication equation is as follows:

$$\begin{aligned} \rho_i P &= pmk_{v_i} + r_i h_i + r_i C_i P \\ &= U_i K_{pri} + \mu_i + r_i h_i + r_i C_i P \\ &= U_i K_{pub} + PID_{v_{i1}} + h_i S_i + C_i S_i \end{aligned}$$

4.2 Security proof using random oracle model

In this subsection, we use the random oracle model to prove the security proof of the unforgeability of our proposed scheme. First, we briefly introduced the Random Oracle Model, analyzed the rationality and background of the proof method, and finally, made proofs for different attackers.

4.2.1 Security model overview

In 1993, Bellare and Rogaway formally proposed the Random Oracle Model (ROM) methodology [68], which led to rapid and significant progress in the field of practical applications of the previously purely theoretical provable security methodology, as well as the creation of "concrete security or exact security" which means that we can obtain a more accurate measure of security precisely. The theory of practically oriented provable security has been widely accepted by academia and industry. Reduction is provable security theory's most commonly used tool or reasoning method. In a provable security scheme, we first established a security model. We built Challenge algorithm C (simulating the challenger role) to prove the scheme's security. Algorithm C reduces a complex security problem to a mathematically complicated problem (e.g., extensive number decomposition, solving discrete logarithms, CDH problems, etc.). The reduction and inference in ROM are generally expressed by first formally defining the security of the scheme and assuming that an adversary can break the scheme's security with non-negligible probability (e.g., forged signatures) within

probabilistic polynomial time (PPT). Then, challenger C provides the adversary with a simulated environment (ROM) that is indistinguishable from the real environment and answers all the adversary's oracle inquiries (the simulated adversary can obtain all attack conditions); finally, use the results of the adversary's attack (such as the presence of a forged signature) trying to solve mathematical problems. In this case, Algorithm C has to embed a random instance of the complicated problem into the adversary simulation process. Still, Challenger C does not know the private key involved in the complex problem. To simulate answering all the adversary's inquiries or to cover up the fact of not knowing the private key (i.e., to simulate a real attack environment) so that the adversary thinks he is interacting with a real challenger, Algorithm C needs some unique ability to compensate for the ignorance of the private key. This ability is realized through a random oracle during the construction process. In the proof process, the hash function is regarded as a random function, that is, a random oracle. Anyone can obtain the value of the hash function by asking the random oracle so that by controlling the oracle, C can exploit the adversary's ability to solve mathematical problems while providing a realistic attack environment. The proof model obtained using the above oracle properties is called the random oracle model.

4.2.2 Security Proof Overview

In the VANETS system, It is necessary to protect information sharing between vehicles and vehicles or between groups and other groups. To prove the security of our proposed scheme. According to [59, 66, 69, 70], a standard security-proof method is to use interactive games with attackers for disproof. The core idea of this method is first to present a known intractable problem (usually a mathematical problem). Then, challenger C tries to solve the problem while attacker A tries to break the security of the solution process. Challenger C can use the interaction with attacker A to come up with an answer. If attacker A cannot break the security of the algorithm, then it can prove that the algorithm is secure. So, we consider the following types of opponents and design the game according to their individuality:

1. Type-I Adversary A_1 (outsider)

Generally, it is a third-party attacker other than SEAF and legitimate users. This opponent is also called a public keys replacement attacker, who can destroy the vehicle's private keys or actively change any vehicle's public keys but does not have access to the master private key.

2. Type-II Adversary A_2 (SEAF)

Generally an honest but curious SEAF, this adversary is also known as a malicious SEAF attacker, who can access the master private key located at SEAF but cannot replace any vehicle's public key.

The proposed scheme is proved unforgeable under the random oracle model [71] for the above two types of attackers. The random oracle model is a theoretically perfect and completely random hash function type widely used as a proof method in provable security. Challenger C plays the controller role in the interaction, and the opponent provides some random oracle model functions for the attacker. An attacker can train using these functions in an attempt to break the security of the interaction:

a) $H_1\text{-query}(RID_i, \alpha_i)$:

C check whether (RID_i, α_i) exists in L_1 . If it exists, then return h_1 . Otherwise, choose a number $\alpha_i \in \mathbb{Z}_q^*$ at random, record (RID_i, α_i) in L_1 , and return α_i .

b) $H_2\text{-query}(\alpha_i, TID_i)$:

C checks whether (α_i, TID_i, GID_i) exists in L_2 ; if it exists, then return h_2 . If not, then randomly select a number $GID_i \in \mathbb{Z}_q^*$, record (α_i, TID_i, GID_i) in L_2 , and return GID_i .

c) $H_3\text{-query}(\alpha_i, TID_i)$:

C check whether (PID_i, U_i) exists in L_3 ; if it exists, then return U_i . If not, then randomly select a number $U_i \in \mathbb{Z}_q^*$, record (PID_i, U_i) in L_3 , and return U_i .

d) $H_4\text{-query}(PID_i, M_i, h_i)$:

C check whether (PID_i, M_i, h_i) exists in L_4 ; if it exists, then return h_i . If not, choose a number $h_i \in \mathbb{Z}_q^*$ at random, record (PID_i, M_i, h_i) in L_4 , and return h_i .

e) $Extract\text{-query}(PID_i, pmk_{(v_i)})$:

C checks whether $(PID_i, PID_{(v_{i1})}, pmk_{(v_i)})$ exists in Ex_list , if it exists, it returns $(PID_{i1}, pmk_{(v_i)})$, otherwise, if $PID_i \neq PID$, C uses the list of L_3 to get the corresponding parameters (PID_i, U_i) . C randomly selects a number $\mu_i \in \mathbb{Z}_q^*$ and calculates PID_{i1} and pmk_{v_i} in Ex_list . Then, C randomly chooses a number $\mu_i \in \mathbb{Z}_q^*$ and calculates PID_{i1} and pmk_{v_i} , and records them in Ex_list $(PID_i, PID_{i1}, pmk_{v_i})$. Scenario 2 $PID_i = PID$ C terminated process.

f) $sign\text{-query}(PID_i, M_i)$:

If $PID_i \neq PID$, C terminates the process. Conversely, if $PID_i = PID$, C obtains the required parameters (PID_i, U_i) and (PID_i, M_i, h_i) for the calculation using L_3 and L_4 . C randomly selects $\rho_i, ri \in Z_q^*$ and calculates S_i and $PID_{v_{i1}} = \rho_i P - U_i K_{pri} - h_i S_i$. Next, C transmits $(\sigma_i = \{\rho_i, S_i\}, PID_i)$ to the adversary.

g) **Forgery**(σ_i):

When using **sign-query** to obtain a valid, legitimate signature $\sigma_i = \{\rho_i, S_i\}$, then according to the forking lemma mentioned in [72], when $PID_i^* = PID_i$, C can obtain another forged valid signature $\sigma_i^* = \{\rho_i^*, S_i\}$.

Definition: If it is impossible to find any polynomial-time **Type 1** and **Type 2** opponents who can win **Game 1** and **Game 2** with a clear advantage in the VANET. Then, the authentication scheme is called provable Security, which is unforgeable.

4.2.3 Unforgeability

In this subsection, we demonstrate the unforgeability of the proposed protocol through interactive games with **Game I** and **Game II**.

• Game I

In this subsection, we demonstrate the unforgeability of the proposed protocol through an interactive game with a Type-I adversary and Challenger C.

Assume that there is a polynomial adversary. A_1 with a non-negligible cracking probability $\xi > 0$, while a forger who can break the proposed security by **H_1 - query**, **H_2 - query**, **H_3 - query**, **H_4 - query**, and **sign-query** can forge a valid signature, Challenger C can then use the interaction with the adversary. A_2 to solve the ECDLP puzzle:

A. Setup

C sets the master public key to $K_{pub} = Q = kP$, initializes the system parameters, and publishes them to the opponent A_1 . So that he can call them. The system parameters are $(G, q, P, K_{pub}, P_{pub}, H_1, H_2, H_3, H_4)$, where (H_1, H_2, H_3, H_4) is a random oracle controlled by C.

B. Training Phase

Adversary A_1 can perform the following multiple polynomial-limited queries on C for learning. In the interaction with A_1 , C maintains L_1, L_2, L_3, L_4 , these initially empty lists.

1. **H_1 -query**(RID_i, α_i)

2. **H_2 -query**(α_i, TID_i)
3. **H_3 -query**(PID_i, U_i)
4. **H_4 -query**(PID_i, M_i)
5. **sign-query**(PID_i, M_i)
6. **Forgery**(σ_i)

C. Forgery

If opponent A_1 generates a forged signature σ_i^* and wins the game as PID_i^* on message M, C uses PID_i^* as A_1 's target identity on message M.

1. Randomly generate $TID_i^* \in Z_q^*$, and calculate $\sigma_i^* = H_1(RID_i, \alpha_i)$ and $GID = H_2(\alpha_i, TID_i)$.
2. Randomly choose a number $\rho_i^*, r_i^* \in Z_q^*$ and calculate $S_i = r_i^* P$ and $PID_{v_{i1}} = \rho_i^* P - U_i K_{pri} - h_i S_i$.
3. Set $\sigma_i^* = \{\rho_i^*, S_i\}$.
4. Return σ_i^* and GID .

Moreover, the σ_i generated by **sign-query** is valid because σ_i satisfies.

$$\begin{aligned} \rho_i P &= U_i K_{pub} + PID_{v_{i1}} + h_i S_i \\ &= U_i K_{pub} + (\rho_i^* P - U_i K_{pri} - h_i S_i) + h_i S_i = \rho_i^* P \end{aligned}$$

The forged σ_i^* generated by **Forgery** can also satisfy.

$$\rho_i^* P = U_i^* K_{pub} + PID_{v_{i1}} + h_i S_i$$

According to the above two formulas, we can calculate

$$(\rho_i - \rho_i^*)P = (U_i - U_i^*)xP$$

Then, by the above equation, we can calculate the discrete logarithm problem $x = (\rho_i - \rho_i^*) / (U_i - U_i^*)$. Therefore, challenger C can solve the ECDLP problem by solving for x . Let qSQ represent the number of query Sign_query oracle. So, we can calculate that the probability that Challenger C can solve the ECDLP problem is at least $1/qSQ(1 - 1/qSQ)^{qSQ-1}\xi$.

• Game II

In this subsection, we demonstrate the unforgeability of the proposed protocol through an interactive game with a Type-II adversary and Challenger C. Assume that there is a polynomial adversary A_2 with a non-negligible cracking probability $\xi > 0$. In contrast, a forger who can break the proposed security by **H_1 - query**, **H_2 - query**, **H_3 - query**, **H_4 - query**, **Extract - query**, and **sign-query** can forge a valid signature. Challenger C can then use the interaction with the adversary, A_2 to solve the ECDLP puzzle:

A. Setup

C sets the master public key to $K_{pub}=Q=kP$, initializes the system parameters, and publishes them to the opponent, A_2 so that he can call them, the system parameters are $(G, q, P, K_{pub}, P_{pub}, H_1, H_2, H_3, H_4)$, where (H_1, H_2, H_3, H_4) is a random oracle controlled by C.

B. Training Phase

Adversary A_2 can perform the following multiple polynomial-limited queries on C for learning. In the interaction with A_2 , C will maintain $L_1, L_2, L_3, L_4, Ex_list$, these initially empty lists.

1. $H_1\text{-query}(RID, \alpha_i)$
2. $H_2\text{-query}(\alpha_i, TID_i)$
3. $H_3\text{-query}(PID_i, U_i)$
4. $H_4\text{-query}(PID_i, M_i)$
5. $Extract\text{-query}(PID_i, pmk_{v_i})$
6. $sign\text{-query}(PID_i, M_i)$
7. $Forgery(\sigma_i)$

C. Forgery

If opponent A_2 generates a forged signature α_i^* and wins the game as PID_i^* on message M, C uses PID_i^* as A_2 's target identity on message M.

1. Randomly generate $TID_i^* \in Z_q^*$, and calculate $\alpha_i^* = H_1(RID_i, \alpha_i)$ and $GID = H_2(\alpha_i, TID_i)$.
2. Randomly choose a number $\varrho_i^*, r_i^* \in Z_q^*$ and calculate $PID_{i1} = \mu_i P$ and $pmk_{v_i} = U_i K_{pri} + \mu_i$.
3. If $PID_i = PID^*$ is obtained in the result of **Extract - query** calculation, then the challenger C aborts the simulation. Otherwise, the process continues.
4. Randomly choose a number $\varrho_i^*, r_i^* \in Z_q^*$ and calculate $S_i = r_i^* P$ and $PI_{v,1} = \varrho_i^* P - U_i K_{pri} - h_i S_i$.
5. Set $\alpha_i^* = \{\varrho_i^*, S_i\}$.
6. Return α_i^* and GID

Moreover, the σ_i generated by **sign-query** is valid because σ_i also satisfies.

$$\begin{aligned} \varrho_i P &= U_i K_{pub} + PID_{v,1} + h_i S_i \\ &= U_i K_{pub} + (\varrho_i P - U_i K_{pri} - h_i S_i) + h_i S_i \\ &= \varrho_i P \end{aligned}$$

The forged σ_i^* generated by **Forgery** can also satisfy.

$$\varrho_i^* P = U_i K_{pub} + PID_{v,1} + h_i^* S_i$$

According to the above two formulas, we can calculate

$$(\varrho_i - \varrho_i^*)P = (h_i - h_i^*)yP$$

Then, by the above equation, we can calculate the discrete logarithm problem $y = (\varrho_i - \varrho_i^*)/(h_i - h_i^*)$. Therefore, challenger C can solve the ECDLP problem by solving for y. Let q_{EX} to represent the number of query **Extract - query** oracle. So, we can calculate that the probability that Challenger C can solve the ECDLP problem is at least $1/q_{EX}(1 - 1/q_{EX})^{q_{EX}-1}\xi$

Although our group leader's idea and ROM proof refer to the previous paper in [66], since BATKA mainly uses the existing cryptographic ECC to generate the required key, we do not need to prove its confidentiality but focus on the forgery-proof. Therefore, we have different kinds of query functions and fewer numbers than [66].

4.2.4 Security Goals Analysis

In this subsection, we analyze the security attributes of our design for the authentication process in an in-vehicle mobile network and the resistance to several common attacks. We also integrate our design to achieve the analyzed security goals.

1. Message authentication and integrity:

The recipient can use our proposed authentication process when receiving a message certificate byte. Since h_i is involved in the message to calculate the hash value, when the message changes, it causes h_i to change, thus causing the authentication to fail so it can ensure the authentication and integrity of the message.

2. Privacy-preserving:

Since the pseudo-identity of the vehicle does not contain real information when it is computed, an attacker cannot get any real information from the pseudo-identity of the vehicle.

3. Traceability:

Once the attacker registers as a legitimate vehicle and intentionally spreads false information, the AMF can calculate $\alpha_i = PID_{v,2} \oplus H_2(\mu_i P_{pub} \| T_{v_i})$ and query the data list with the obtained α_i . Based on the RID found, AMF puts it on the revocation list.

4. Replay attack:

Since the transmitted message contains the timestamp T_i and is used to generate the signature σ_i , where h_i is also

included in the timestamp T_i , the recipient can check and verify whether T_i is expired, thus defending against replay attacks.

5. Forging attack:

According to **Game I** and **Game II**, attackers are equivalent to the need to solve ECDLP. ECDLP is mathematically a complicated problem. An attacker cannot solve it in polynomial time.

4.3 ProVerif

4.3.1 Overview

ProVerif [73] is a formal verification tool based on the Dolev-Yao model for automatic verification of security protocols. It supports cryptographic primitives, such as symmetric encryption, public key cryptography, key exchange protocols, etc. ProVerif first converts the protocol into a Horn clause and checks for security by automatic inference, enabling vulnerabilities to be discovered during the protocol design phase. When a vulnerability is found, a corresponding attack sequence is output.

4.3.2 Validation process and result

To use ProVerif for protocol verification, it is necessary to convert the protocol into a format that ProVerif can read. We write our proposed authentication scheme into a.pv file used by ProVerif so that ProVerif can validate BATKA, and the process of writing it is described below.

In the file, we classify the system parameters into public or non-public-type assumptions according to their nature. Then, define the channels the devices used to transmit, the associated RIDs and SUPIs, and the functions defining each key and generating the hash values. Finally, we define the initial part of the system for our proposed authentication scheme based on these defined parameters and functions as inputs. Subroutines define the various actions involved in the authentication between the vehicle and the vehicle, including calculation and decryption, transmission of messages, etc. We included some system parameters and subroutines in the main program section to start the process. Then, we define

the attack sequence for the desired test security items and let ProVerif know that these items need to be verified and checked. ProVerif then runs the files we write to validate the checks against the security items we define and outputs the results.

The ProVerif verification results of our proposed authentication scheme are shown in Fig. 6. The items we tested are two scenarios: the vehicle identity authentication of the group leader by the vehicle and the identity authentication of the vehicle by the group leader. The results in Fig. 6 show that the validation results are safe.

5 Performance evaluation

In this section, we analyze and compare our proposed schemes. First, we individually derive the computational overhead for single message and batch authentication through theoretical analysis and compare it with similar schemes. Then, we simulate the proposed authentication process with the Python PBC suite to validate our theoretical analysis. In this way, we demonstrate the effectiveness and advantages of BATKA. Since the various batch authentication scenarios and the components used to generate the authentication parameters are different, the overhead and transmission generated are also different, so the analysis in this section focuses only on the authentication overhead and the authentication performance at different vehicle counts.

5.1 Theoretical evaluation

In this subsection, the authentication process of our proposed scheme is compared with theoretical analysis to discuss computational and storage overheads.

5.1.1 Computational overhead

In this subsection, the symbols used for the calculation overhead of our theoretical analysis are as follows:

T_{ECC-m} Calculate the time spent on multiplying groups of elliptic curves.

Fig. 6 ProVerif results

```
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Verification summary:
Query event(termV(x,y)) ==> event(acceptGLV(x,y)) is true.
Query inj-event(termGLV(x,y)) ==> inj-event(acceptV(x,y)) is true.
-----
```


Table 5 Computational overhead

	Single authentication	Batch authentication
BATKA	$3T_{ECC-m} + 2T_{ECC-a} + 3T_h$	$(n+2)T_{ECC-m} + nT_{ECC-sm} + nT_{ECC-a} + 3nT_h$
EMBA [55]	$3T_{ECC-m} + T_{bp} + T_h$	$4nT_{ECC-m} + T_{bp} + nT_h$
ECAS [56]	$2T_{ECC-m} + 4T_{bp} + 2T_H + T_h$	$(4n-4)T_{ECC-a} + 4nT_{ECC-m} + 4T_{bp} + 2T_H + nT_h$
CM-CPPA [61]	$3T_{chev} + 2T_{h-chech}$	$3nT_{chev} + 2nT_{h-chech}$
ANAA-Fog [62]	$3T_{ECC-m} + T_{ECC-a} + 2T_h$	$(n+2)T_{ECC-m} + (n-1)T_{ECC-a} + 2nT_h$
ECA-VFog [63]	$2T_{ECC-m} + 2T_{ECC-a} + T_h$	$(n+1)T_{ECC-m} + (3n-1)T_{ECC-a} + nT_h$

T_{ECC-sm}	Calculate the time spent on small-scale multiplying groups of elliptic curves.
T_{ECC-a}	Calculate the time spent in the addition group of elliptic curves.
T_{chev}	Calculate the time spent on Chebyshev's polynomial mapping operation.
T_{h-chev}	Calculate the time spent on the chaotic map hashing operations.
T_h	The time it takes to execute the one-way hash function.
T_H	The time it takes to execute the map-to-point hash function.
T_{bp}	The time is taken to perform the bilinear pairing operation.

Single authentication applies when handover; we use the group leader as the group representative to perform the group handover. Another scenario is when there is no need to process so many messages simultaneously. First, we present the proposed and other schemes'calculated expenses, as shown in Table 5.

5.1.2 Storage cost overhead

This subsection analyzes the storage cost overhead of the proposed scheme and compares it with existing schemes. To achieve a security level of at least 80 bits of security, we use 160 bits of ECC for communication cost comparison since 160-bit ECC provides the same level of security as that provided by the 1024-bit RSA public key cryptosystem [57]. In elliptic curves, the length of a point on the Elliptic Curve is usually expressed as 320 bits or 40 bytes. On the other hand, the length of the point in the bilinear mapping is 128 bytes, and the element used as the private key in Z_q^* should be 20 bytes. Finally, the required length of the timestamp should be

4 bytes. We refer to Table 6 in [74] and summarize our related parameters, as shown in Table 6.

The parameters that need to be broadcast during the batch authentication phase are $\{GID, M_1, PID_1, \sigma_1, T_1\}$, requiring a bandwidth of $2 \times 40 + 3 \times 20 + 2 \times 4 = 148$ bytes. The communication costs of other schemes, along with BATKA, are summarized in Table 7. Table 7 shows that Anil Kumar Sutrala's scheme has the least storage size, while ours is the second largest.

5.2 Simulation

We use Python to simulate the computational overhead spent on the authentication process of BATKA and other schemes and compare them to each other to present our results.

5.2.1 Overview

Pairing-Based Cryptography library (PBC) [75] is an open-source pairing-based cryptography library developed by a team of researchers at Stanford University through long-term research and practice. This PBC library is designed

Table 6 Bits in bilinear pairing and ECC

Type of the System	Pairing	Length of elements of the group	$ G $
ECC	Without Pairing	$ G = 320bits$	$q = 160bits$
Bilinear Pairing	$e : G_1 \times G_1 \rightarrow G_T$	$ G_1 = 1024bits$	$q = 160bits$

Table 7 Comparison of communication cost

scheme	Storage message	Communication costs
BATKA	$\{PID_{i1}, PID_{i2}, T_i\}, GID, \sigma_1, T_1$	148 bytes
EMBA [55]	$P_{ui}, C, Sig, s_1, s_2, s_3$	220 bytes
ECAS [56]	$msg_i, t_i, vepk_i, PSUID_i, ID_{R_i}, sig_i$	172 bytes
CM-CPPA [61]	$PID_{vj}, M_i, T_{vi}, \sigma_{vi}$	44 bytes
ANAA-Fog [62]	$Msg_i, PAID_{i1}, PAID_{i2}, ts_i, \sigma_i$	108 bytes
ECA-VFog [63]	$m_{v,i}, AID_v, R_{pub}, D_{v,i}, T_{v,i}, \sigma_{v,i}$	124 bytes

Table 8 Cryptographic operations and execution times

	Times (ms)
T_{ECC-m}	8.0940
T_{chev}	3.8878
T_{h-chev}	0.7023
T_{ECC-sm}	0.6109
T_{ECC-a}	0.0372
T_h	0.0022
T_H	17.1117
T_{bp}	19.9421

to provide an efficient and secure paired-base encryption scheme with a wide range of applications. The library offers powerful tools and functions for implementing pairing-based cryptographic algorithms and related cryptographic operations. The library is constantly updated with the latest pairing-based cryptographic theories and algorithms. It can be used in different programming languages, such as Java and Python, making verifying and designing cryptography-related schemes easier for researchers.

5.2.2 Simulation result

In this subsection, to simulate and evaluate the time cost of the calculations for the above theoretical analysis, we chose to use the open-source python-based pairing-based cryptography (pypbc) library [76] for Class A calculations, and the simulation environment, we installed the pypbc suite on an Asus's desktop computer (i7- 11700 4.90 GHz) in VMware's virtual machine (8 processors, 8 GB RAM, Ubuntu 20.04) and calculated the time consumed by the above eight computations shown in Table 8. Finally, the time spent on authentication is calculated using the theoretical analysis mentioned in Section 5.1.1.

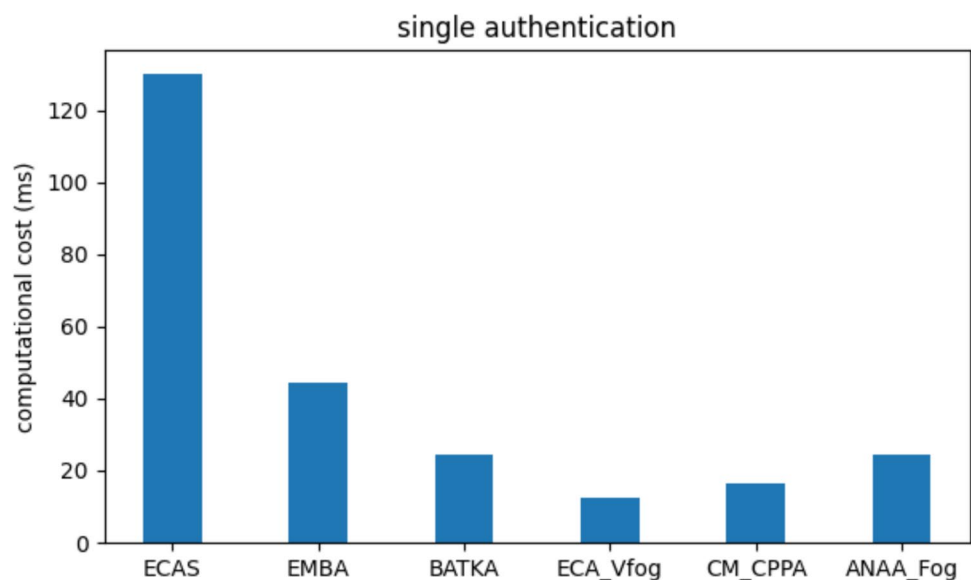
Fig. 7 Time cost of single authentication

Figure 7 shows that, regarding the computational cost for single message authentication, the ECAS scheme takes the longest, followed by the EMBA scheme. The proposed BATKA scheme has a lower computational cost than ECAS and EMBA. Still, it is higher than CM-CPPA [61], ANAA-Fog [62], and ECA-VFog [63].

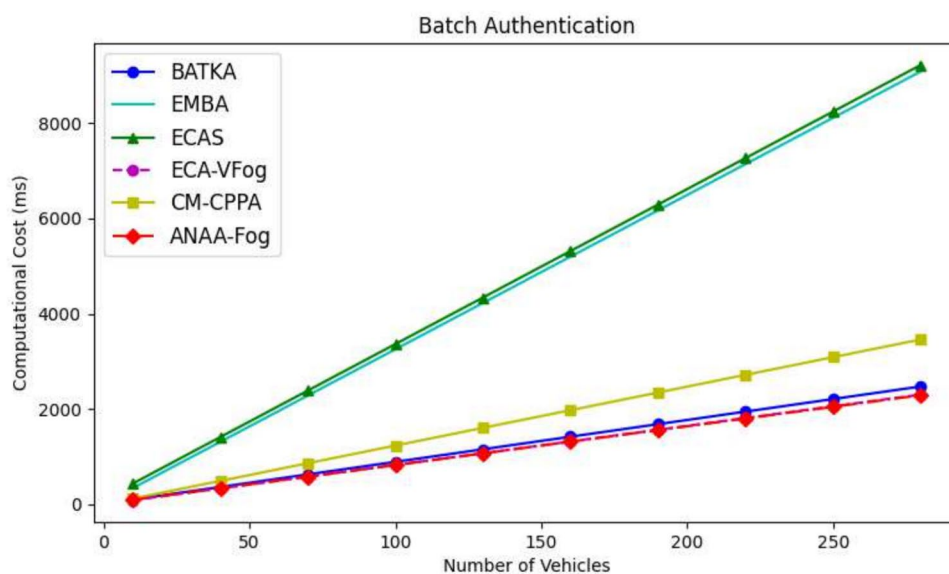
As shown in Fig. 8, both EMBA and ECAS schemes utilize bilinear pairing operations, but their time overhead does not significantly increase with the number of authentications. We speculate this is because the bilinear pairing operation contributes the most to the overall authentication time compared to other cryptographic operations.

Additionally, although Chebyshev's polynomial mapping operation has a lower computation time than elliptic curve multiplication, BATKA only requires a single elliptic curve multiplication. As a result, BATKA still maintains an advantage over CM-CPPA in terms of time efficiency. Furthermore, while BATKA is slightly slower than ECA-VFog and ANAA-Fog, its performance is nearly identical. Moreover, BATKA incorporates additional optimizations within the 5G core network authentication process, making it a valuable approach despite its slightly higher computation cost.

6 Conclusion

In the future, smart vehicles become more and more popular, and mobile networks develop rapidly. According to the vehicle's mobile nature, the vehicle's network communication function mainly uses the network function provided by the mobile network. The high-speed mobility of vehicles reduces the computational overhead of authentication. Coupled with the original user volume of the mobile network, adding new vehicle users at this time causes the 5G infrastructure

Fig. 8 Time cost of batch authentication



to handle a larger number of users. Therefore, to achieve anonymous mutual authentication among vehicle users and reduce the massive computational overhead of connection authentication for 5G infrastructure or RSU, we design a batch authentication process for vehicle groups and combine tree-based Key agreement used in 5G and VANET scenarios.

We first propose a secure batch authentication scheme based on ECC by combining the pseudo-ID and SUPI commonly used for identification in VANET. This scheme authenticates with 5G services, so AMF is responsible for vehicle registration. SEAF then generates a pseudo-identity and a partial private key for its vehicles, which can be updated periodically according to security requirements. Then, in the VANET message delivery, information can be used by BATKA to facilitate the vehicle's authentication of information. At the same time, the vehicle's identity privacy protection is through the VANET information authentication, which is done through a fake identity authentication rather than using the actual identity, reducing the association with the real identity so that users can reduce the risk of being traced. At the same time, this privacy protection is conditional. If a malicious user successfully registers properly, this privacy protection protects the behavior of malicious attacks. So, they should be responsible for the registration of AMF in addition to any device or vehicle that can not be associated with the real identity from a fake identity. BATKA has been proven secure through the random oracle model, demonstrating its unforgeability and comprehensively analyzing its necessary security properties and resistance to various potential attacks. Finally, through theoretical analysis and simulation, BATKA was validated to be effective in reducing the computational effort and performing well in batch authentication compared with others.

In future vehicular networks, satisfying various needs and providing high-quality services are important research goals while considering information security and computing efficiency. In the future, we move towards edge computing with lower latency or with the development of multi-factor authentication. It is expected that the authentication of the vehicle network can be combined with multi-factor authentication and added to FIDO Alliance so that biometric factors can be introduced into the design of future authentication schemes to achieve influential associations between vehicles and users. Before the user starts the vehicle, the user is identified by the user's biometrics, and only legitimate users can enter and start the vehicle.

Acknowledgements Taiwan National Science and Technology Council partly sponsored this work under contract NSTC 113-2221-E-194-041-. In addition, we would like to thank the editors and the anonymous reviewers for their valuable comments and suggestions. Their efforts have significantly improved the quality of this article.

Author contributions Conceptualization [W.-K. Chiang]; Methodology: [W.-K. Chiang]; Performance analysis and investigation: [P.-Y. Liao, C.-J. Liu]; Writing—original draft preparation: P.-Y. Liao, C.-J. Liu; Writing—review and editing: [W.-K. Chiang, C.-J. Liu]; Funding acquisition: [W.-K. Chiang]; Supervision: W.-K. Chiang].

Funding W.-K. Chiang received funding from Taiwan National Science and Technology Council under contract NSTC 113-2221-E-194-041- for the submitted work.

Data availability No datasets were generated or analysed during the current study.

Declarations

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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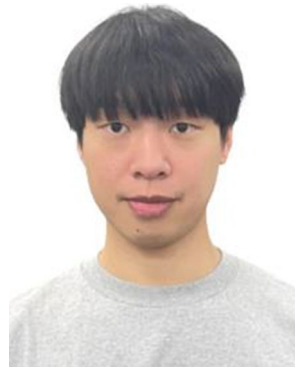


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